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ORIGINAL RESEARCH PAPER

LoRa based intelligent soil and weather condition monitoring with internet of things for precision agriculture in smart cities

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Abstract

Urbanization is expected to hold about 50% of the world population by 2050 and there will be stress on available resources including food and freshwater. Further, inefficient utilization of irrigation water possesses additional challenges and it is predicted that by 2050, India will face shortage of freshwater resources. Urban agriculture and Precision Agriculture are the possible solutions in smart cities for the predicted challenges providing food security and safety, sustainable life in cities. Internet of Things (IoT) and Machine Learning (ML) are the possible solutions to smart cities' farming. Agricultural activities such as irrigation of plants, rain and drought conditions monitoring supported by IoT and ML succour farmers to righteous decision on agriculture activities. A detailed study is carried out on the role, opportunities, and different aspects of smart cities, urban farming, communication technologies, IoT and ML with respect to agriculture. The article presents the design of intelligent irrigation system based on soil and weather conditions. The soil and weather parameters are selected through various research articles in Agriculture 4.0 and ML. The article also juxtapositions the designed weather station with the various patents developed. The system developed in this article provides a cost-effective and state-of-the-art solution to local weather monitoring.

1 | INTRODUCTION

Agriculture is the use of land for the cultivation of crops and breeding of animals in a created ecosystem to provide food, medicinal plants and other products to sustain and enhance life. Nowadays, agricultural practices are not only limited to rural areas. Due to urbanization and shift of population towards cities, urban agriculture is providing food security and sustainable life in cities. As the cities are becoming smart cities, smart solution for urban agriculture is also required. Urban agriculture includes all agricultural activities like breeding animal, fertilization and most importantly irrigation. Urban agriculture includes all the sited for food production including the green spaces of gardens, universities and home plantation. Proper irrigation practices are required at these sites for efficient utilization of irrigation water. The trend of urbanization in recent years [1, 2] has put much stress on agriculture due to loss of land and labor. Urban farming is one of the solutions to provide food and lessen load on rural agriculture and provide locally grown food to citizens and reduce cost of transportation [3]. Urban agriculture includes aquaculture, livestock, and plants newly developed concept of agriculture, that is, urban agriculture applies to both rural and urban areas [4].

Precision Agriculture is the use of technology in agriculture so as to provide better solution to monitoring and controlling of agriculture parameters and thus improving overall production of agriculture products. It involves monitoring of plants, crops, soil and weather parameters, livestock etc. Precision Agriculture helps in efficient utilization of agricultural resources. For example, irrigation, an important aspect of Precision Agriculture and plays an important role in agricultural sector. About 40% of the global irrigation water is supplied by ground water and in India around 39 million hectares of land is irrigated by ground water. India only 4% of world's fresh water resources, out of which

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80% is used for agriculture. So, the efficient utilization of irrigation water through Precision Agriculture is very crucial for profitable and sustainable agriculture practices [5, 6].

Monitoring of soil, weather and plants is important for efficient usage of agricultural resources. The important components of agricultural monitoring are: data collection sensors, data transfer using IoT modules, IoT cloud services, application server for data analysis and monitoring device [7]. IoT provides the advanced way to make devices smarter and enable then to communicate and share the information. IoT has been developing by utilizing the various technologies and being deployed by several applications. The IoT application areas and henceforth the number of devoices connected through IoT are rising exponentially. According to various IT reports, IoT market is going to increase by $14.4 trillion by 2022. IoT implementation works with the objective of providing simple end user interface for data visualization, mor and more deployment of web enable devices and lowering the barrier in connecting things to one another. Wi-Fi is the main technology behind the IoT and Wi-Fi based WSNs are the core of IoT. ML typically aims to learn from past experience, that is, training data set and once the training is completed is capable of classifying or predicting things based on training and unknown dataset [8–11]. LPWAN is being increasingly used in IoT applications. LPWAN offers the low power solution for IoT applications and is important as sensing nodes being deployed in open fields. With the advancement in LPWAN, many promising technologies are offered to agriculture IoT like NB-IoT, Sigfox, Long Range Wide Area Netwok-LoRaWAN. LoRaWAN offers good service for IoT solutions to Precision Agriculture due to its long-range communication at low power consumption. LPWAN LoRaWAN technology has been utilized for monitoring of various agriculture parameter like moisture, humidity, temperature, pH [12, 13]. ML and IoT are the most advanced and proposed ideas in this paper. ML is a special way of teaching the system or machine to learn things and take a wise decision later based on the learnt data. The ML has a list of algorithms and classifiers that works based on the situation and data. For a better efficiency of system, one has to choose appropriate algorithm and classifier. There lie the ways in improving the model performance. The collection of data and transmission of data is done through IoT technologies. Connecting the devices through internet and the communication is held over internet in most of the cases in this new era. IoT has brought a great impact on society making people understand the need of technology in different fields. The IoT and ML have combined today in most of the fields in giving a valuable and worthy solutions. The weather station and data transmission system is done using IoT and prediction system is done using ML in the proposed idea, which gives a strength and value to the invention on technology point of view. Comparative study of different weather stations based on the pre-requisites of the smart irrigation is elaborated. The irrigation scheduling based on ML is the motive of the paper. The paper is intended to find the list of parameters that an innovative irrigation system needs. The weather station is built with the parameters and the value is logged into cloud platform (ThingSpeak).

1.1 | Motivation of this paper

IoT and ML are the leading technologies of the time. Many of the areas like health care, image processing, Wireless Sensor Networks (WSN), cyber and network security are getting benefitted from these technologies. The need of improvement in the Precise Water management, due to inefficient conventional irrigation practices, has forced the authors to study and work on various technologies available to propose a unique effective solution. The Agriculture 4.0 motivates the researchers to propose different futuristic ideas based on IoT and ML to improve the sector as the need increases day by day. The IoT based Weather station and ML based prediction system to initiate a IoT based irrigation system would cause an effective change in implementation is predicted and the work is carried out. The various research questions intended to be explored thought the present article are

- RQ1: What is urban farming and role of urban farming in smart cities
- RQ2: What are the primary soil and weather conditions affecting the irrigating planning, and prediction?
- RQ3: What are commercially available solutions to monitoring soil and local weather conditions?
- RQ4. What are the different agricultural aspects adapting ML actively?
- RQ5. What role does ML play in water management?
- RQ5. Are the commercially available solutions to monitoring local weather conditions feasible for deployment by the farmers?

2 | RELATED WORKS

Smart cities are the topic of interest amongst researchers and development of IoT is providing many opportunities in smart cities. But on of the aspect in smart cities is being neglected is urban farming and minimal research in available on urban farming. This section explores the state-of-the-art literature for smart cities, Urban farming, IoT and use of in agriculture.

2.1 | Smart cities

The major supporting element for smart cities is communication mainly wireless communication technologies like information system, monitoring of devices, and control over offered services, etc. The importance of Information and Communication Technology (ICT) is elaborated and stressed in many of the literature [4, 14–17]. In [18–20], smart cities have been defined as:

A smart city can be described as an urban locality that employs digital data and technology to create efficiencies for boosting the economic development, enhancing the quality of life, and improving the sustainability of a city

Smart cities have many definitions, ranging from those focusing on the technical aspects like what technologies are important

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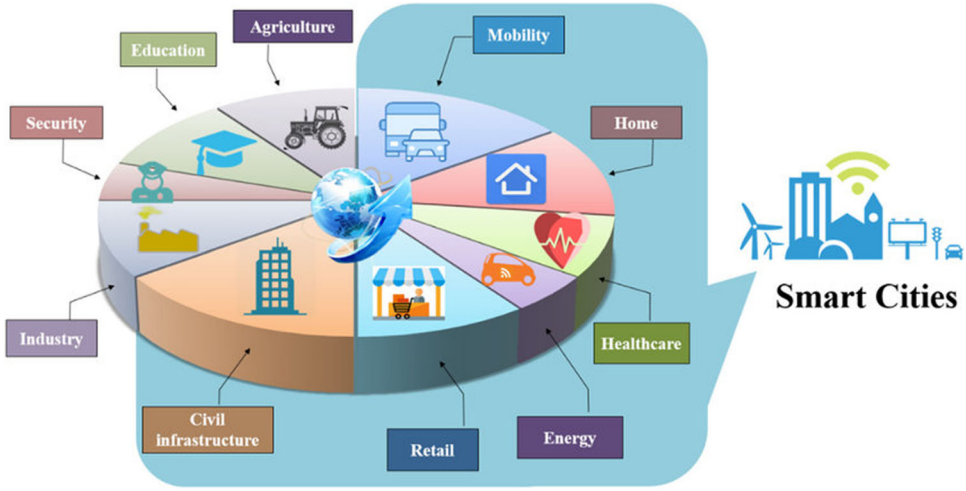


FIGURE 1 Smart city application areas

to those focusing on social aspects like enabling citizens and communities to act smarter

A smart sustainable city is an innovative city that uses information and communication technologies (ICTs) and other means to improve quality of life, the efficiency of urban operations and services, and competitiveness, while ensuring that it meets the needs of present and future generations concerning economic, social and environmental aspects.

The conception of smart cities includes the city organization and management as a whole through technology. It is constructed and maintained by using integrated technologies, such as sensors, electronics, and networks.

There are many aspect of smart cities where smart solution are possible like education, homes, manufacturing, transportation, farming etc. Each application area of smart city is o associated with many of the sub-aspects and are elaborated in Figure 1 adapted from [21–24].

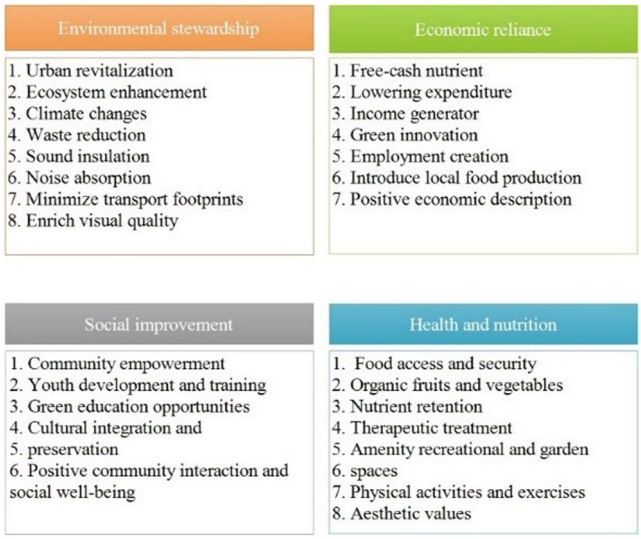


FIGURE 2 Urban farming benefits

2.2 | Urban agriculture and IoT

Urban farming is d or livestock growth or process that are undertaken within an urban a or around the urban centers with the purpose to generate income, as defined in [25]. Urban farming is possible on rooftops, in city grounds, and also on vertical walls. Three types of urban farming have been identified namely community farming, vertical farming, and rooftop farming. Urban farming in addition to providing food and source of income also contributes towards the city's green space, garden for recreational purpose and conserve the environment and ecology of the city. The four potential benefits of urban agriculture have been identified and elaborated in Figure 2 [25].

IoT provided the mechanism for devices to communicate with each other and makes the devices smarter. IoT is developed by utilizing various technologies like Embedded Systems, Wireless Sensor Networks (WSN), communication technologies, cloud computing and hence being deployed in many applications. The application areas and thereby the number of devices utilizing IoT are steeply increasing [26]. Industrialization and intensification of agriculture is required to improve the quality and quality of agricultural products to meet increased demand and efficient utilization of agriculture resources. IoT provides propitious solutions for agriculture modernization by providing remote methods for agricultural monitoring and control. Agriculture application using IoT are livestock management, precision farming and greenhouse management. Different WSN wireless technologies used for agriculture monitoring are Zigbee [27–31], LoRaWAN [26, 27, 32–36], Sigfox, NB-IoT [27, 33, 37], Wi-Fi [27, 38–42]. Bluetooth [27, 38, 43–45], RFID [27, 29, 46], nRF24l01 [27, 28]. Farm land with IoT based sensors and Wireless Sensors Networks (WSNs) gathers relevant information and communicates it to framers. Some of the IoT setups for agriculture do analysis the data and suggestions are made to farmer for better planning of agriculture related activities. IoT implementations work with the objective of giving end user

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TABLE 1 Data and range capacity of LPWAN technologies

Communication technology	Range	Data rate
Bluetooth	Short	Low
BLE	Short	Low
Zigbee	Short	Low
6LowPan	Short	Low
NFC/RFID	Short	Low
LPWAN	Long	Low
VSAT	Long	Low
Wi-Fi vLow	High	
5G	Moderate	High
4G	Moderate	High
3G	Moderate	High
2G	Moderate	High

simple interface for data visualization, more and more deployment of web-enabled devices, and lowering the barrier in connecting things to one another. Wi-Fi based WSNs node are the important elements in IoT technology. Different radio communication technologies are classified on the basis on their range (short, long, moderate) and data rate (low, high), as summarized in Table 1. [11, 47].

WSN is gaining much popularity mainly due to development of IoT based applications and 5G systems. In IoT infrastructure, WSN is considered to be as one of the main contributing elements. With the continuous demand and penetration of WSN in different applications, mainly due to steep rise in IoT enabled devices, power efficient and effective communication between sensor and network sink is required. There are many WSN parameters which contributes towards the effective deployment of WSN technology such as signal strength, access technologies, battery life, energy consumption. In WSN, energy consumption can be improved through scheduled data transmission and by using direction antennas [48]. Power efficient and long range WSN solutions are also required in Precision Agriculture as the sensor nodes, to gather information about soil and weather conditions, are installed at very remote locations and frequent service or requirement of frequent replacement of battery will not be feasible. Zigbee, Bluetooth, WiFi and GPRS/3G/4G technology, Long Range Radio (LoRa) protocol, Sigfox are the various wireless protocols available and can be exploited for Precision Agriculture. Zigbee is considered as one of the best members for Precision Agriculture (PA) wireless sensor networks. Zigbee provides communication range up to 100 m and as minimum as 30 m for indoor applications. Zigbee wireless protocol is adopted in many applications such as water quality management, irrigation automation, greenhouse monitoring and livestock monitoring. Bluetooth is used for the short-range communication upto 10 m and with BLE range can be extended up to 100 m. Bluetooth technology is being used in monitoring weather, soil moisture, temperature and sprinkler positioning using Global Positioning System (GPS).

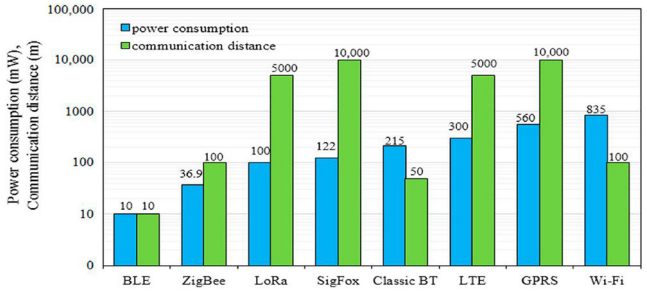


FIGURE 3 Urban farming benefits

In Precision Agriculture, WiFi protocol was used to collect agriculture data like soil moisture, soil temperature, air temperature and humidity, sunlight intensity, Carbon Dioxide. The agricultural information so obtained is sent to a gateway for storage before being transmitted to server system through Wi-Fi network. Global Packet Radio Service (GPRS) is GSM – based data packet service and with WSN is utilized for automatic crop irrigation system. A prototype for drip irrigation system was developed and evaluated for soil moisture measurement. For transmitting the soil, plant and weather information GPRS was equipped with various wireless sensor nodes. WSN - GPRS forms a gateway between WSN and GSM for transmitting the information from WSN to data management centre where further analysis of data can be done using mobile phones, tablets or computers. SigFox is a wireless cellular network for low data rate applications and was used to develop geo location system to localize animals in mountain pastures during summer. A system to locate cattle and enhance their productivity was implemented using SigFox. Next and the latest contender in WSN is LoRa alliance established in 2015 and Low PowerWide Area Network -LoRaWAN is a low power, wide area IoT communication protocol. LoRa network basically consists of LoRa end devices, LoRa gateway and LoRa network service. LoRa end devices is used for communicating with LoRa gateway which employs LoRa with LoRaWAN. The raw LoRaWAN data packets are passed to LoRa network server by LoRa gateway. LoRa protocol was utilised in monitoring of soil moisture, soil temperature, air temperature, air humidity and light intensity inside greenhouse using different sensors and microcontrollers. Further comparing the various WSN protocols, as shown in Figure 3, LoRa is found as one of the most power efficient solution for long distant communication. By using Wireless Sensor Network Monitoring system in precision irrigation, the water saving of 90% water using Zigbee [49], 33% of water saving using Zigbee [50], 90% of water saving using Bluetooth [51] is achieved in comparison to tradition irrigation systems [37, 52–62, 65] (see Figure 3).

2.3 | Machine learning in agriculture 4.0

A novel Collaborative Filtering (CF) method algorithm is proposed in the literature [35] and is claimed to be outperformed the conventional algorithms for Deep Learning based collaborative filtering algorithms. CF is used in various electronic

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commerce application like YouTube, Amazon for recommendation systems. User preference is predicted by CF after observing the preference history of other similar users. There are two classes of CF

- Nearest neighbor methods which predict the user preferences by considering the rating given by some other similar users.
- Model based method developed with ML and uses training data to find the pattern for prediction used preference.

The proposed CF in [66] is based on supervisor machine meaning technique. Learning of the machine can be done through supervised or unsupervised methods. Supervise ML method requires a distinct training data but in unsupervised ML there is no difference between test and training data. Some of the learning models not limited to are regression, Clustering, Bayesian, Instance based, Decision tree and Artificial Neural Network. ML has been used in different fields of agriculture as shown in Figure 1 [8].

Learning of machine is an iterative process and performance of system under training ameliorate over the time. Once the training is completed the model so trained can be employed for classification, prediction and creating new testing data. Typical ML approach consists of training dataset fed to ML algorithm for training. Once the training is completed, the trained algorithm is deployed for prediction and classification of unknown data. Digital agriculture has emerged as a new field aiming to improve the conventional less efficient agricultural practices, most importantly irrigation. Pressure on agriculture will be increasing as a result on continuous increase in world population. Agriculture utilizes technology called precision agriculture or also termed as digital agriculture. For efficient utilization of agriculture resources like fertilizers, pesticides and most importantly irrigation water new methods need to be adapted. Now machine leaning in agriculture has provided opportunities for understanding the data intensive processes related to livestock, soil, crop and water management involved with agricultural environment. ML provided the machines with the capability to learn. Researches and literatures punished reveals that ML has been used in Precision Agriculture for soil management, livestock management, crop management and water management. ML algorithms use for soil management, livestock management, crop management and water management in precision Agriculture is given in Figure 4.

Water management or precision irrigation is of paramount importance as fresh water resources in the world are limited and less efficient conventional irrigation methods lead to wastage of irrigation water along with change in soil properties and reduction in crop yield quality and quality. ML with IoT not only provides the proper monitoring and control tools for irrigation but also assists farmers in irrigation planning by predicting the crop water demand for proper irrigation. Figure 5 give the different ML algorithms used for water management in agriculture based on parameters evapotranspiration and dew point temperature, as shown in Figure 5. For water management the important weather parameters measures are temperature, humidity and wind conditions [8, 9, 28, 45, 67–70].

Agriculture withdraws 70% of the worldwide water and Precision Agriculture is an effective tool for efficient utilization of irrigation water. The advantage of Precision Agriculture is profitability, good productivity, precise selection of fertilizers, decrease chemical bills and fuel cost and systemically structured irrigation planning. Remote sensing is a low cost and less labor-intensive strategy for farm monitoring. Remote sensing is helping farmers in various applications like classification of crops, forecasting crop yield, disease detection and management, harvest arrangement and crop water stress detection. ML is an effective tool for classification of systems. By applying ML to remote sensing data, a real time farm management solution to farmers can be provided. The article [9] demonstrate that the irrigation scheduling based on crop water stress management not only uses the irrigation water efficiently but also improves the agriculture yield. Various applications of ML algorithms used for water stress determination is reviewed. Various methods for crop water stress measurement as highlighted in the article are shown in Figure 6.

The article has also provided the important ML algorithms which are employed in agriculture and summarized in Figure 6. Support vector machine, Random Forest classifier, eXtreme gradient boosting (XGBoost), Rotation Forest, Oblique random forest and Artificial Neural Networks are some of the widely used ML algorithms for crop water stress management in agriculture. Weather monitoring is the new area of interest amongst the researchers due to global climatic changes. Global climatic changes like drought and heavy rain are also affecting the utilization of irrigation water. Evapotranspiration is an important factor in determining the irrigation demand or in irrigation scheduling. Sensing evapotranspiration using remote sensing devices and estimating evapotranspiration using ML to the received data offers an efficient way for irrigation planning. For estimating evapotranspiration, it is important to gather weather temperature, humidity and wind condition information. Weather station is a setup, used to collect weather data and provides the information to the user. The classification of weather station present in various literatures is provided in Table 2. The table has classified the weather monitoring devices on the basis of the way data is collected, transmission technology, deployment of ML and irrigation scheduling. Most of the researches/literatures have considered evapotranspiration prediction using ML for irrigation scheduling [9, 71].

2.4 | Study on various irrigation models

The ignition system of irrigation is automated through IoT and the decision taken using ML algorithms. The prediction is done using ML or different process. Different models are developed for irrigation scheduling by using weather station and ML algorithms. The weather station models developed in different researches/literatures is discussed in Table 3. The most of the ML algorithms consider Temperature, wind speed, wind direction, humidity weather conditions.

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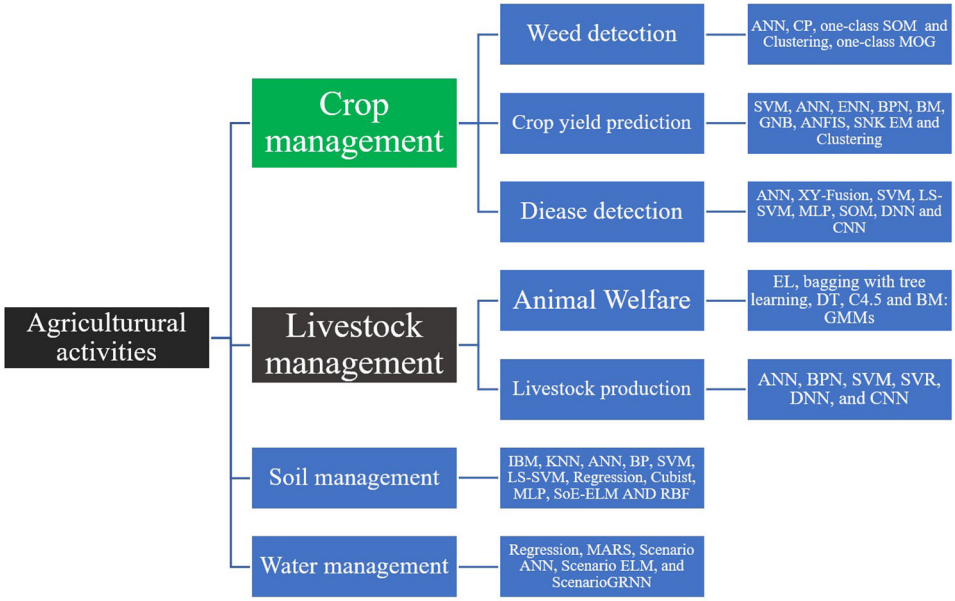


FIGURE 4 ML algorithms deployed for agricultural application

TABLE 2 Classification of weather station

Parts	Functionalities	References	Key points	Conclusion
Data collection	Weather stations are built or data is taken from some online site and fed to the system.	[72–88] [89]	Data collection from Sensor node is proposed. The station is proposed to be Automatic Weather Station-AWS where the system sends the data to remote user.	Data collection is done from a single node in most of the times. The need of collection from multiple nodes for processing makes the system efficient.
Transmission technology	The transmission of data from the sensor node to the central hub.	[72, 74] [77–81] [90, 91]	NB IoT, Thing speak, Fog and Edge computing, Zigbee, Lora, Wi-Fi, cloud, email, twitter are different ways proposed.	The efficient use of different systems was proposed. The work on Lo-Ra is not widely seen in the proposed application.
ML	Predicting the functionality based on the received data	[73, 80, 84] [91–94]	ML based predictions were done in a better way. The prediction accuracy was mentioned to be above 90 in these papers.	Building a system with feature with a real time data is proposed earlier. The actuator and sensor system connected to a single ML based system where the actuator is actuated using ML decision is not widely seen in the research.
Irrigation scheduling	Scheduling the irrigation based on ML results and initiating IoT services to enable irrigation system.	[70, 79, 90] [91–99]	The irrigation system includes of actuator based on the IoT signal is proposed and, in some cases, the decision from pre-loaded data is data is fed as a control signal.	The real-time data that is used to evoke the ML system to predict and actuate the system is not seen in most of the studies. The complexity of merging up the technologies in giving an efficient solution isn't seen in most of the cases.

TABLE 3 ML algorithms applied to weather datas

Year/reference	Weather station measured parameters	ML technique for irrigation scheduling
2015 [72]	Temperature, humidity, wind velocity, atmospheric pressure, rain	NEN and ANN
2020 [92]	Wind speed, wind direction, temperature, rain, humidity, light	TCN
2020 [93]	Temperature, humidity, solar radiation, wind speed	Boost regressing model
2020 [98]	Based on soil conditions	VAR and RNN-LSTM
2019 [99]	Air temperature, humidity, wind speed, precipitation, Radiation	Fuzzy logic

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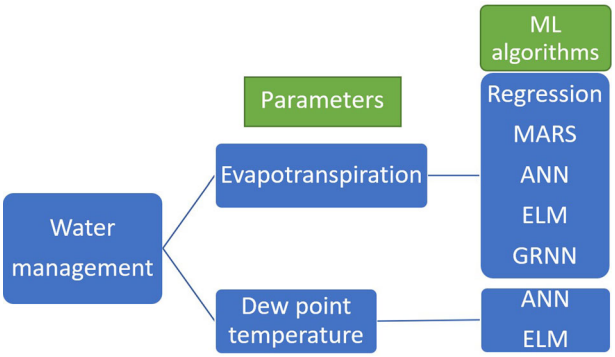


FIGURE 5 ML algorithms deployed for agricultural application

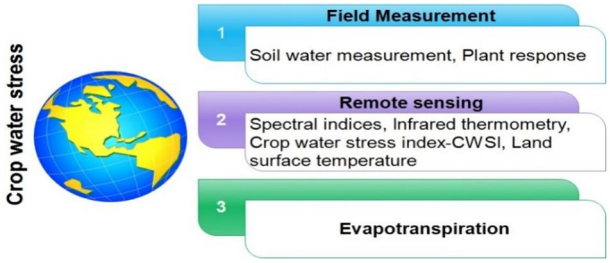


FIGURE 6 Crop water stress measurement methods



FIGURE 8 Soil monitoring sensor node

3 | SOIL AND WEATHER MONITORING SYSTEM

The various element of urban farming is elaborated in Figure 7 [2, 25 84]. The proposed system for soil and weather condition monitoring consists of two modules namely LoRa based soil sensor node and Weather monitoring local weather station. Weather station is connected to weather station through LoRa wireless communication module and to IoT cloud. Weather station uploads the soil and weather conditions to IoT cloud for monitoring and assisting in irrigation scheduling.

3.1 | Soil monitoring

The important soil parameters to be monitored for irrigation planning are soil temperature and soil moisture. Soil moisture estimation is the direct way of irrigation scheduling and

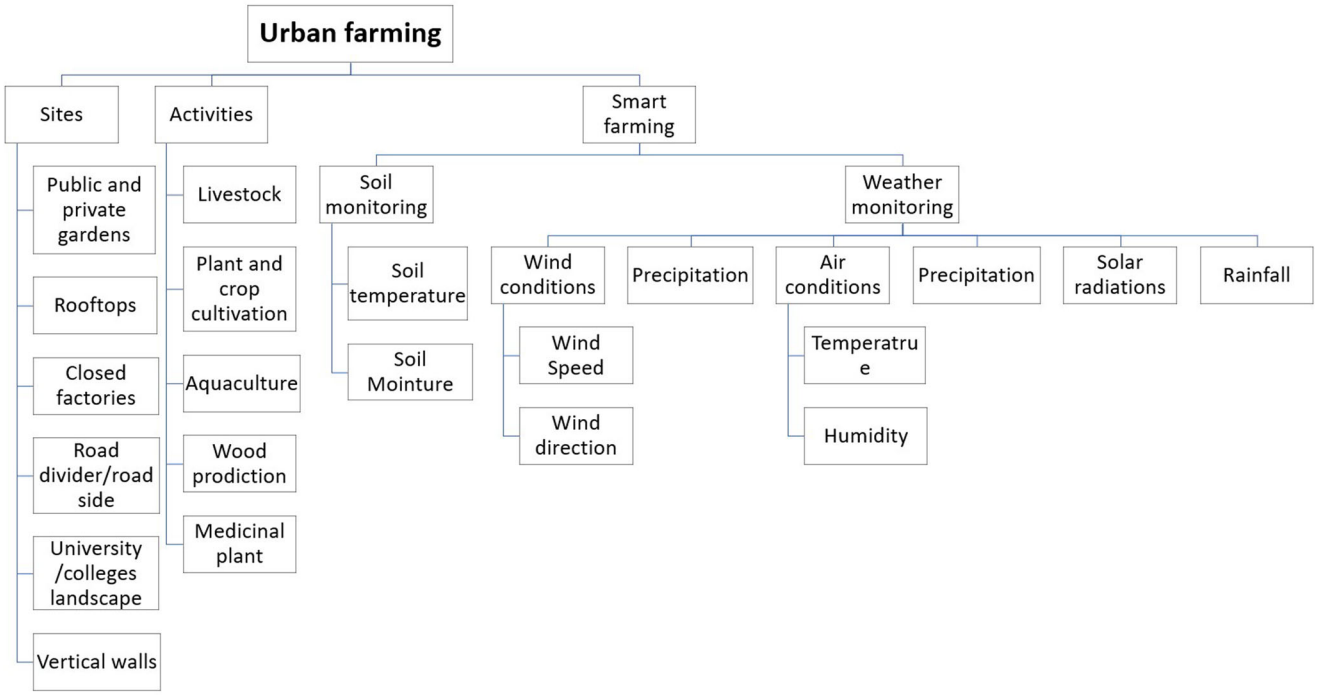


FIGURE 7 Components of urban farming

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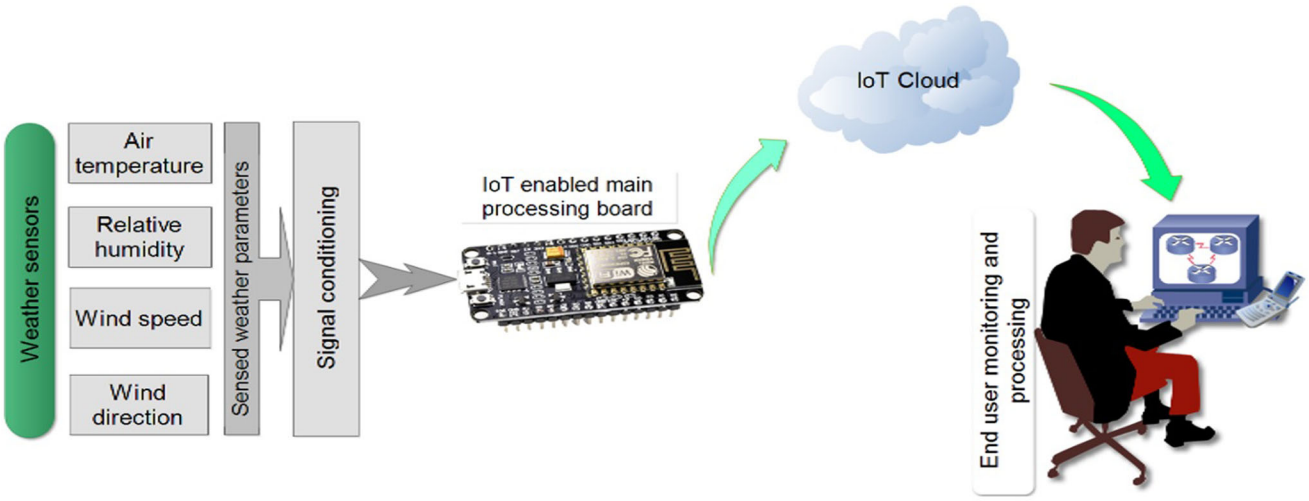


FIGURE 9 Block diagram of developed system

temperature of soil helps in predicting the irrigation requirement. The soil condition monitoring smart sensor node consist of temperature sensor, soil moisture sensor, processing unit and LoRa module for communication. Figure 8 shows the onsite deployed sensor node. Soil sensor node provides the soil condition sense data to weather station.

3.2 | Weather station design

Weather station is a provision, equipped with sensors for measuring environmental conditions. Already mentioned, due to climatic changes environmental monitoring is becoming an active research area. Climatic conditions do affect and influence the irrigation planning. Evaluation and advances in IoT have created new paradigm for environmental monitoring. IoT has provided the possible solution for expansion of current weather monitoring infrastructure [37]. After evaluating various researches and literatures for irrigation planning based on weather conditions, the weather parameters affecting the irrigation scheduling and its prediction using ML are:

- wind speed,
- wind direction,
- air temperature and
- humidity.

The weather conditions to be sensed by the weather station are specifically selected to aid in utilizing ML algorithms for irrigation prediction and scheduling. The weather station developed gathers the required weather information with the help of suitable sensors. The gathered information is provided to the user via IoT. IoT enables the user to monitor field weather condition as and when required without physically visiting the farm land. Weather station developed aims to provide cost effective solution to local weather monitoring by the farmers. The developed weather station is designed to be low cost, for this free IoT cloud server Thingspeak has been selected for data storage and visualization. Figure 9 provides the developed weather station block diagram.



FIGURE 10 Weather station

4 | RESULT AND DISCUSSION

The system so developed is tested and the data is monitored for more than a month in November-December, 2020. The local weather station developed for farm land employees DH22 for gathering and temperature and relative humidity. DHT22 is a digital sensor and is much precise and has more range than its counterpart DHT11. Figure 5 shows the installed weather station for measuring weather conditions and measured weather conditions Thingspeak visualization of air temperature, relative humidity and wind speed is shown through Figures 10–13,

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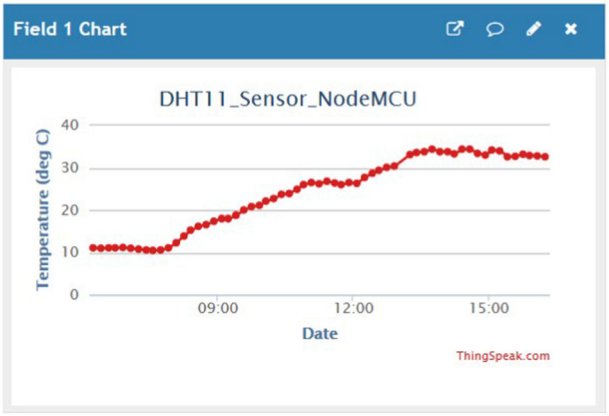


FIGURE 11 Air temperature data visualization

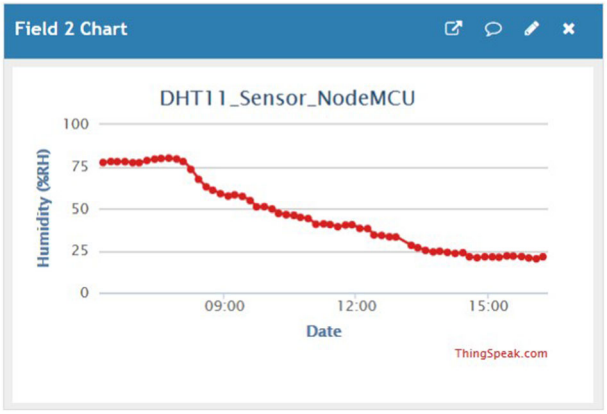


FIGURE 12 Relative humidity data visualization

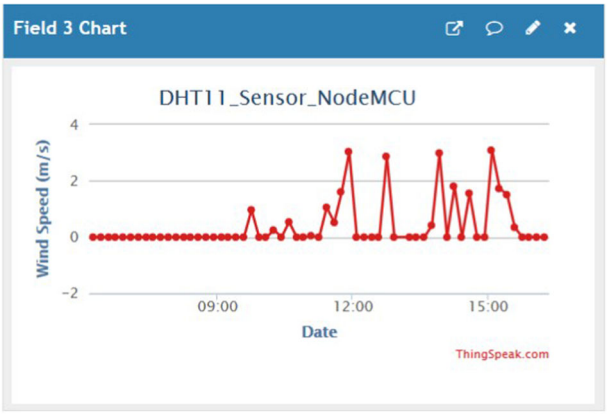


FIGURE 13 Wind speed data visualization

respectively. Figure 14 gives the mobile interface of Thingspeak cloud.

The weather station developed is a cost-effective system for efficient utilization of agricultural irrigation water. Along with the irrigation water efficient utilization, agricultural yield quality and quantity can also be increased through predictive irrigation by applying the ML algorithms to the provided weather information. The system developed is much cost effective (less than 50% of the commercially available weather stations without IoT) with IoT technology in comparison to most of the commercially available weather station. The commercially available weather station lacks in IoT technology. They either use local display or GSM for data visualization and transmission that too at heavy cost addition. Table 4 provides the cost comparison (in INR) of different commercially available weather stations.

Normally the commercial weather stations start from 70,000/- and hence quite high cost is involved in the development of commercially available weather station. Moreover, the commercially available solutions for weather monitoring are expensive, require specific Information and Communication (ICT) skills to operate, extensive infrastructure and resources. This makes the commercially available weather stations infeasible for operation by farmers. Owing to high cost and difficulty in operating commercial weather stations, farmers avoid their deployment in farm lands and continues with the convention less efficient agricultural practices. By making them aware of the advantages of deploying the technological solutions in agriculture and providing them accessible low-cost solutions, farmer can be motivated to shift from conventional farming to Agriculture 4.0. Using state-of-the-art IoT technology makes the developed system more reachable; smart and data visualized on Thingspeak cloud helps in irrigation related decision quickly and precisely. Apart from offering advantage in terms of cost, the developed system is adapted with the most suitable weather parameters, selected after thorough and detailed literature survey, which are of paramount importance in irrigation planning, scheduling and prediction via ML. Table 5 compares the developed weather station with the published patents on weather station. The patents so selected are searched using Google patent search engine with the string weather station and weather station WSN.

5 | CONCLUSIONS

Role of IoT, remote sensing, ML in agriculture is explored. The papers summarize the various WSN technologies for IoT, agriculture applications using IoT and ML. The article also discussed the importance of soil and weather conditions in irrigation panning and environmental conditions affecting the irrigation planning the most. IoT with ML assist farmers in planning agriculture activities precisely. Different ML algorithms employed in soil monitoring, livestock management, crop and water management are summarized. The literature review summarizes the various methods for crop water stress estimation and also discloses that evapotranspiration estimation plays a vital role in crop water stress management. Irrigation planning based on crop water stress helps in planning irrigation affectively. In the end the article presents the design of IoT enabled weather station with test results. The developed weather station is compared with the commercially available weather station and the patent available on weather station for its novelty.

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TABLE 4 Cost comparison of commercial weather stations

S. no.	Weather station product	Cost/unit	Make	IoT feature
1	Automatic weather Station, for industrial: ATM1136[a]	70,000	Advance Tec India Private Limited	No
2	V Tech ultrasonic weather station for agriculture: VT- UWS01[b]	120,000	V TECH	No
3	Portable weather station: 110-WS-18[c]	325,000	Auro Electronics Private Limited	No
4	Automatic weather station, usage: Industrial: RK900-01[d]	143,000	Infinity Enterprise Private Limited	No
5	Automatic weather Station, for official[e]	500,000	Nevco Engineer Private Limited	No

[a]https://www.indiamart.com/proddetail/automatic-weather-station-12477572848.html
[b]https://www.indiamart.com/proddetail/ultrasonic-weather-station-19054403862.html
[c]https://www.indiamart.com/proddetail/portable-weather-station-4539296991.html
[d]https://www.indiamart.com/proddetail/automatic-weather-station-18244123933.html
[e]https://www.indiamart.com/proddetail/automatic-weather-station-13513226588.html

TABLE 5 Developed weather station comparison with patents

Patent ID	Patent Title	Year	Description	Developed System
US10638675B2	Irrigation controller with weather station	2020	The invention is irrigation controller based on local weather station data	The present system is weather station design with IoT technology. The weather parameter is such that can help in predicting irrigation demand by using ML.
CN203465825U	Farmland environment weather information collection system based on Zigbee and GPRS	2014	The invention provides the information about soil moisture, soil temperature, wind speed, wind direction, illumination intensity, Carbon-di-oxide and air temperature fo the farmland under observation. Zigbee and GPRS communication technology is used for data transmission to terminal server.	Discussed invention uses Zigbee, suitable for short distance communication only and GPRS which is a very infrastructure-oriented resource. The weather data of farmland is not user friendly. The developed system is a centralized weather station, uses latest IoT technology for data transmission to cloud. IoT enables the user to visualize the data in different formats like text or graphs fr analysis manually to tools like ML.
US7171308B2	Weather station	2007	Weather information provided by present invention are thunderstorm and lightning strike.	The discussed system provides the weather information like wind speed, wind direction, humidity and air temperature for irrigation planning. The developed weather station used IoT technology and data is analysis ready for advanced tool like ML.
US7088221B2	Weather station	2006	The invention discusses the weather station design with microprocessor unit, atmospheric sensors, memory to store weather data and local display unit.	The developed system is IoT based with no local display. The data available on cloud can be utilized for prediction of irrigation requirement with advanced ML tools.
US6967900B2	Combination clock radio, weather station and message organizer	2005	Weather station in present invention consist of temperature sensor, pressure sensor and local display for data visualization.	The weather system developed consists of sensor required to monitor weather conditions for irrigation scheduling. System is IoT enable not requiring any local memory for data storage and ML like advance analysis tool can be applied to data set from weather station for irrigation demand prediction.
US5920827A	Wireless weather station	1999	Weather parameters are measure and transmitted using radio transmitter. Also used local display for data display.	The system developed uses IoT technology enabling user to visualize data as and when required. Also, irrigation prediction (using Machen Learning) along with irrigation planning is possible with available data from the developed weather station

Due to the climatic change worldwide, environmental monitoring is day by day becoming one of the active research areas. The commercially available IoT sensors for weather monitoring are very costly and complex. IoT low-cost sensors can be used for weather monitoring but there are certain concerns in deploying the low-cost sensor like frequent calibration requirement and their reliability, in field applicability and performance. For providing reliable and accurate data low-cost sensors requires to be calibrated frequently as they are affected by environmental conditions. Studies has

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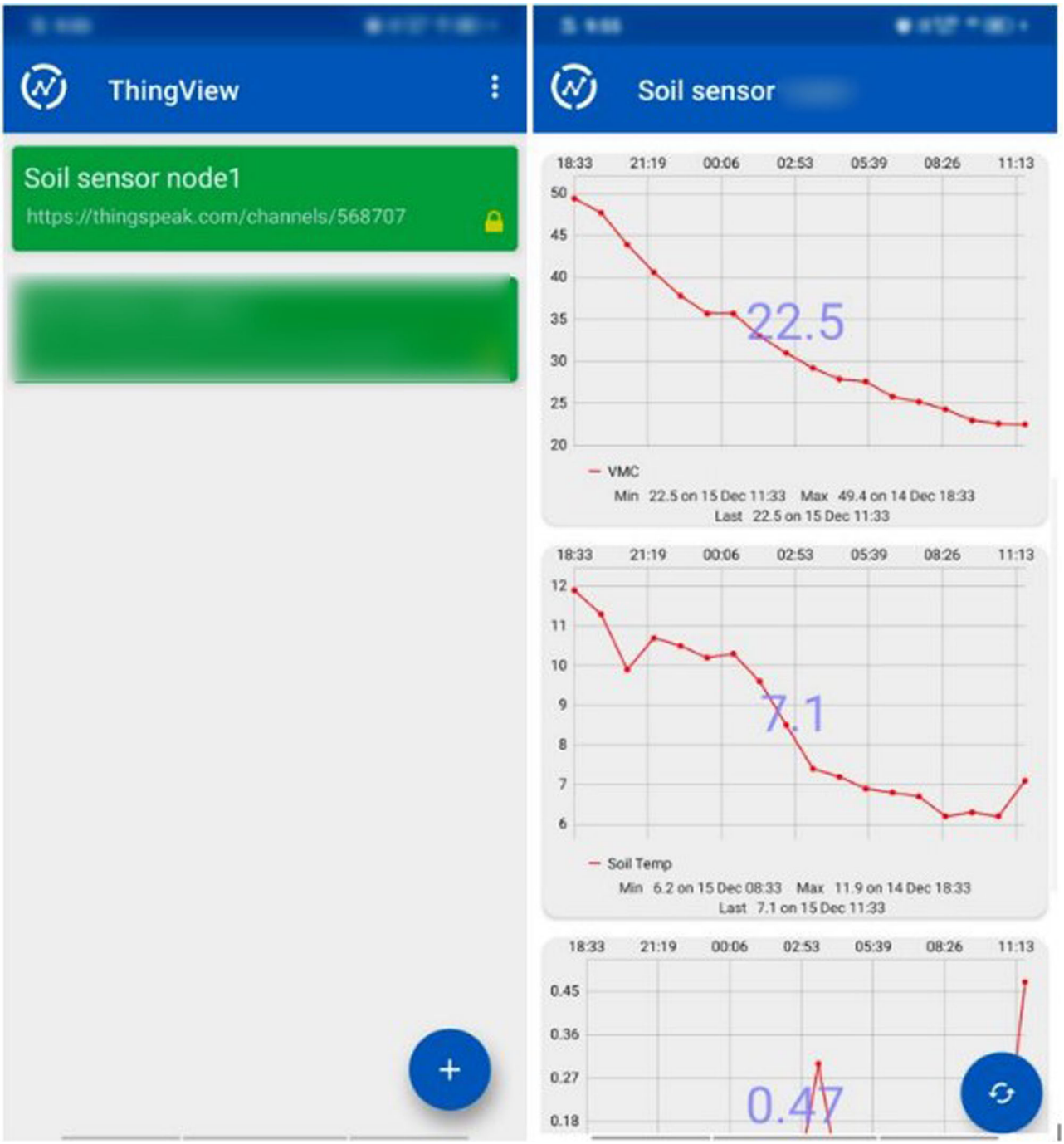


FIGURE 14 Mobile IoT interface

confirmed that IoT low-cost sensors are affected by environmental condition and for calibration of such sensor it is necessary to

- Determine environmental factors affecting that data quality of sensor
- Model the those factor that effects sensor data and
- Apply the model to sensor data for its correction

For modeling of environmental factor affecting the sensor output and applying those models for data correction require ML tools. This study has been carried out to understand the environmental factors contributing to the data quality of low-cost sensors in IoT. In the study performance of FS, FFS, BE, and EFS is evaluated in identifying the factors affecting the sensor data quality [71]. In [94] the challenges related to Precision Irrigation and the identified opportunities for the

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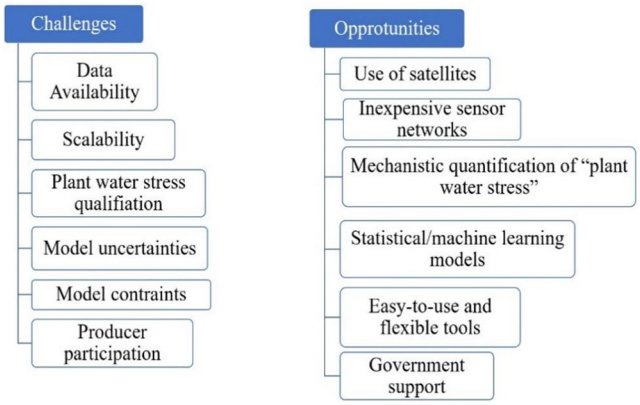


FIGURE 15 Precision irrigation challenges and opportunities

mentioned challenges are elaborated and is summarized in Figure 15.

LIST OF ABBREVIATIONS

6 LoWPAN	Low -Power Wireless Personal Area Networks
ANN	Artificial Neural Network
ANN	Artificial Neural Network
BE	Backward Elimination
BLE	Bluetooth Low Energy
CP	Counter Propagation
CWSI	Crop Water Stress Index
DNN	Deep Neural Networks
DT	Decision Trees
EFS	Exhaustive Feature Selection
ELM	Extreme Learning Machine
FFS	Forward Feature Selection
FS	Feature Selection
GRNN	Generalized Regression Neural Network
ICT	Information and Communication Technology
IoT	Internet of Things
LPWAN	Low-Power Wide-Area Network
MARS	Multivariate Adaptive Regression Splines
NFC	Near Field Communication
RFID	Radio-Frequency Identification
SVM	Support Vector Machines
VSAT	Very Small Aperture Terminal

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CONFLICT OF INTEREST

All the authors hereby declare no conflict of interest of any type

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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